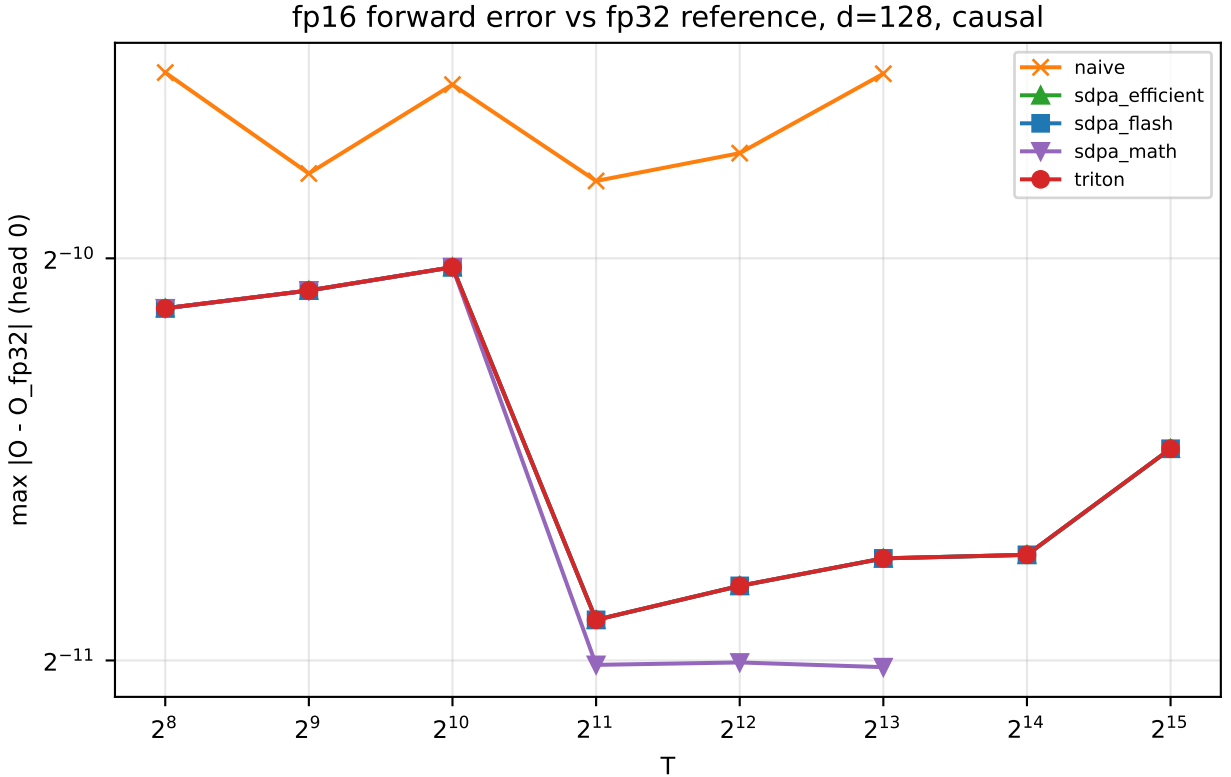
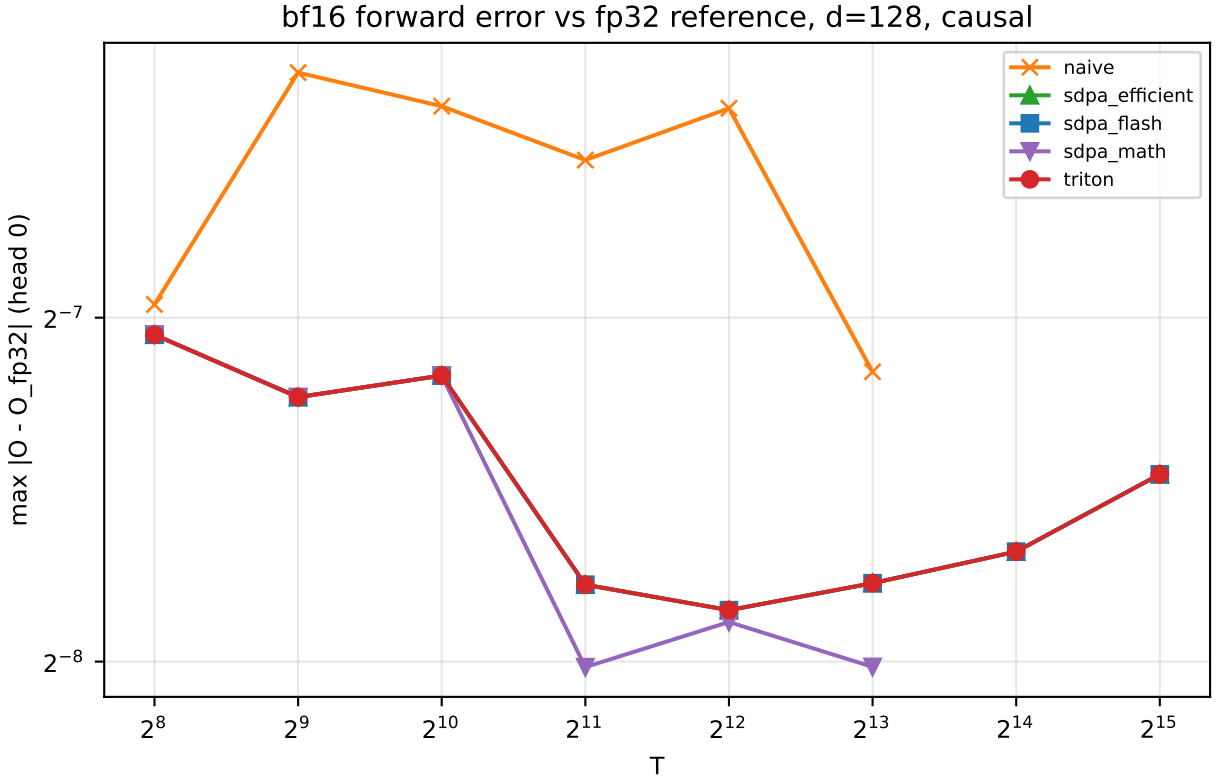

fp16 forward error vs fp32 reference, d=128, causal
max |O - O_fp32| (head 0)
2^{-10}
2^{-11}
2^{8}
2^{9}
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
2^{15}
T
naive
sdpa_efficient
sdpa_flash
sdpa_math
triton


bf16 forward error vs fp32 reference, d=128, causal
max |O - O_fp32| (head 0)
2^{-7}
2^{-8}
2^{8}
2^{9}
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
2^{15}
T
naive
sdpa_efficient
sdpa_flash
sdpa_math
triton